Jyoti Prakash Sahoo

ML Engineer — Python | SQL | NLP | Production RAG | Data Engineering | Azure | PowerBI
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PROFESSIONAL SUMMARY

Machine Learning Engineer with 3+ years building production ML, GenAI, and data pipeline systems at Shell. Delivered 20–35% model performance gains on forecasting/classification workloads, built a Hybrid RAG system improving retrieval effectiveness by 20%, and cut a 2-day manual process to under 5 seconds (480+ hours/year saved). Reduced incident response time and improved system reliability by 90% through anomaly detection on production telemetry. Skilled in Python, SQL, PySpark, LangChain, and Azure.

TECHNICAL SKILLS

- **Languages & Core :** Python (OOP, data structures & algorithms), SQL
- **Machine Learning:** Regression, Classification, Time-Series Forecasting, Bayesian Modelling, MMM, Scikit-learn, Feature Engineering, Cross-Validation,ML Evaluation, Hyperparameter Tuning, ARIMA, SARIMA
- **GenAI / RAG:** LangChain, OpenAI APIs, Hugging Face, Hybrid RAG, Vector Search, BM25, Embeddings, Semantic Search, Prompt Engineering, LLMs, Agentic AI
- **Distributed Systems:** PySpark, Databricks, Distributed Processing
- **Data Engineering:** ETL/ELT, Data Cleaning, Data Validation, Data Transformation, Data Pipelines, ADLS Gen2, Delta Lake, Dimensional Modelling
- **Cloud & DevOps:** Microsoft Azure, Azure Functions, Logic Apps, Azure Data Factory, Key Vault, Entra ID, REST APIs, GitHub Actions, CI/CD, Docker
- **Visualization:** PowerBI

PROFESSIONAL EXPERIENCE

Shell India Markets Pvt. Ltd. *Bengaluru* Aug 2023 – Present
Associate Software Engineer

- **Anomaly Detection:** Engineering an unsupervised Isolation Forest pipeline to detect anomalous Microsoft Fabric capacity states across **137K+ records**, leveraging **10+ temporal** and CU-utilization features to surface emerging throttling risks; implemented using PySpark, Databricks, Python & Power BI *(In Progress)*.
- **ML Modelling- Forecasting & Classification:** Built and evaluated regression, time-series forecasting, and classification models using **Spark ML**, applying feature engineering, hyperparameter tuning, and cross-validation to improve model performance by **20–35%** against baseline approaches.
- **Process Automation (Azure): Single-handedly built** and deployed an end-to-end provisioning automation, from requirements through hypercare, using Azure Logic Apps Workflow with **ServiceNow** integration. It cut provisioning time from **2 days** to **2 seconds** with full audit trails and eliminated **99%** of manual intervention. Also enhanced the legacy event-driven provisioning service (**Azure Functions in PowerShell**, Storage Queues, SharePoint Online, **MS Graph**, **Azure AD**, **IDAPI**), reducing function invocations by **60%**, compute by **40%**, and ServiceNow requests by **20%.**

PROJECTS

- **RAG-Based PCOS Research Assistant - LIVE PROJECT | GitHub | Live Demo :** End-to-end Hybrid RAG system using LangChain, combining BM25 keyword retrieval and vector search with document chunking, embeddings, and LLM generation; improved retrieval effectiveness by **20%** and answer faithfulness by **15%** compared with vector-only retrieval. Evaluated using the **RAGAS** framework — achieved Faithfulness score of **0.82 (82%)** and Answer Relevancy of **0.56 (56%)**, using the gap between them to identify retrieval and prompt-generation improvement areas. . [GitHub]
- **Customer Churn Prediction & MMM***(Scikit-learn, Spark ML):* Classification models (logistic regression, decision trees, random forest) for customer churn prediction, and Marketing Mix Modelling (linear regression, Bayesian, mixed-effects) for driver-based forecasting. [GitHub]

EDUCATION

National Institute of Technology, Rourkela Aug 2019 – Jul 2023
Bachelor of Technology, CGPA: 7.4
Certification: Microsoft Certified: Azure Fundamentals (AZ-900)